

Labwork 8.1 - String Operations

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork8.c**.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (25p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to Labwork 8 String Operations! By using this machine you can do:
1. Swap Characters
2. Trim Unnecessary Parts
3. Count Words
4. Exit
Make your choice:
```

Question 2 - Swap Characters (25p)

- Create a new function (**swap_characters**) that takes a pointer that points the initial character of the string and size of it. (Both the string and the size comes from the main - define the given string below and find out its size by using `strlen()`)
String: `char sentence[] = "The echo of a silent star."`
- In addition to this, you need to take 2 inputs from the user: the indices of the positions that need to be swapped.
- In the end, print the new version of the string.

```
>> Enter the first index: 4
>> Enter the second index: 6
The hceo of a silent star.
```

Question 3 - Trim Unnecessary Parts (25p)

- Create a new function (**trim_plus**) that takes the pointer that shows the initial character of the string and size of it. (Both the string and the size comes from the main - define the given string below and find out its size by using `strlen()`)
String: "a mi+schievous squir+rel cleverly h+id its big+gest acorn++ beneath a pile of freshly fallen ++maple leaves"
- In the end, print the trimmed version of the string.

```
a mischievous squirrel cleverly hid its biggest acorn beneath a pile of
freshly fallen maple leaves
```

Question 4 - Count Words (25p)

- Create a new function (**count_words**) that takes the pointer that shows the initial character of the string, size of it and a key. (Both the string and the size comes from the main - define the given string below and find out its size by using `strlen()`)
String: The mountains are calling and I must go.
- This function will count the words that start with a given key letter and then return it.

```
>> Enter the key: a
There are 2 words that start with a.
```